Zakhar Varfolomeev → ML engineer

About me:

- I am an ML Engineer specializing in the audio domain, with diverse experience developing models for music and speech generation, self-supervised audio models, music transcription, audio tagging, source separation, beat detection, and neural audio codecs.
- Organizer and speaker at conferences on machine learning in Open Data Science, Ontico, etc. communities. I have a lot of public talks about ML for audio.
- I use AI agents wisely and accelerate development.

My contacts:

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- GitHub: @varfolomeeff

Professional experience:

08.2025 - 03.2026

ML engineer (audio processing)

Songsterr

I was working on a very specific and unsolved problem; no one in the world had even come close to solving such a task. Designed and implemented end-to-end R&D for generating fingerstyle arrangements from arbitrary audio. Completely autonomous work without calls and joint discussions.

- Designed and implemented a two-stage ML pipeline for generating fingerstyle arrangements (Audio → MIDI → Guitar Tab) and specialized metrics for evaluating fingerstyle generation outputs. Conducted in-depth domain and data research.
- Developed a MIDI event aggregation pipeline for stage 1 from a lot of tabs across multiple instrument tracks to produce a single melodic line consistent with the full audio mix.
- Collected and cleaned a dataset of ~10,000 original-guitar cover pairs and developed an automatic pipeline for matching audio and tabs.
- Also supervised manual markup by professional arrangers and developed quality and data selection criteria.
- Trained and fine-tuned transcription and generation models (including LoRA, beam search, and structured token masking).
- Conducted a series of architectural and dataset experiments (exploring different audio representations, tokenization strategies, genre-based filtering and so on).
- Achieved production-ready model quality capable of generating valid starting points for fingerstyle arrangements.

04.2025 - 08.2025

ML engineer (audio processing)

Yandex Music

Worked in the generative AI team on models for music generation and music analysis.

- Trained neural audio codecs optimized for music generation tasks.
- Experimented with custom loss functions to improve musical coherence and codebook utilization.
- Applied self-supervised approaches to learn robust and informative audio representations. Trained SSL models from scratch.
- Conducted systematic quality analysis: reconstruction fidelity, robustness to generative noise, and suitability of latent representations for downstream generation models. Created local benchmark.

01.2024 - 01.2025

ML engineer (audio processing)

X-Labs AI

Trained models for generating audio by text description.

- I have integrated into ongoing projects by analyzing and understanding the existing codebase and architecture.
- Then I handled the majority of the development for a large-scale project from scratch. The company uses this in the music generation pipeline.
- Selected data, prepared, and organized the dataset.
- Adapted open-source code to meet the company's specific requirements. Fixed numerous bugs in the codebase, improving system stability and performance.

- Designed the model architecture and developed a roadmap for future improvements. Conducted in-depth research of scientific papers on arXiv to identify optimal approaches and solutions.
- Trained the model using distributed training, efficiently utilizing available computational resources.

08.2021 - 11.2022

Backend developer (python)

FatCode

An online educational platform with WEB development courses for Junior+ level programmers. The project was attracted by the presence of special functionality that other platforms do not have: team dashboard, forum for Q/A, glossary, reward system.

- Designed and wrote a REST API based on Django Rest Framework & FastAPI frameworks for user authorization/registration, creating courses and finding a team to practice after the courses.
- I had a code review and I conducted a code review for others.

My Projects & Publications

2023 - 2026

Audio2MIDI

I am creating an ML model for translating arbitrary audio content into piano notes.

- I create and test hypotheses, work with generative music models based on VQ-VAE & Transformers. I am studying ways to encode audio to extract information useful for piano cover generation.
- Developed four-stage pipline for generating piano arrangments (Form generation from audio → Reduced Lead Sheet Generation → Lead Sheet Generation → Accompaniment Generation).
- Compiled the largest dataset of audio-arrangement pairs, aligned pairs using custom DTW algorithm. Developed heuristics for extracting targets for stages 2-3.
- I made a presentation on the solution at ML conferences.

2026

ESpeech-2

Emotional & On-Device TTS, Dubbing.

- Scaled ESpeech-Pipe to process 1,000,000h of raw audio, yielding a 200,000h clean speech dataset.
- Running experiments on emotional speech synthesis, on-device TTS, and audio dubbing over this corpus.
- Developing a novel GRPO modification for RL post-training of AR TTS (paper in preparation).

2025

ESpeech

Large-Scale Russian TTS Dataset & Pipeline That Better Than Emilia Pipeline.

- Developed ESpeech-Pipe, processing 50,000h of raw audio into 12,801h of clean speech (12,000 speakers, avg MOS 3.3).
- Created SOTA Russian F5-TTS model trained with reinforcement learning.

2025

Automatic codebook size adaptation for retrieval and generation tasks

Artificial Intelligence Research Institute Summer School 2025

- Authored a method for dynamic codebook size adaptation in VQ-VAE via a Codebook Size Predictor (CSP) module with Truncated-GMM sampling.
- Applied to a tokenized-audio LM (Qwen2.5 0.5B): UTMOS improved from ~1.5 to ~2.0 with 99% codebook utilization.